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Review

DNA and chromosomal damage in medical workers exposed to anaesthetic gases assessed by the lymphocyte cytokinesis-block micronucleus (CBMN) assay. A critical review



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ABSTRACT

The lymphocyte cytokinesis-block micronucleus (CBMN) assay has been applied in hundreds of *in vivo* biomonitoring studies of humans exposed either environmentally or occupationally to genotoxic chemicals. However, there is an emerging need to re-evaluate the use of MN and other biomarkers within the lymphocyte CBMN cytome assay as quantitative indicators of exposure to main classes of chemical genotoxins.

The main aim of the present report is to systematically review published studies investigating the use of the lymphocyte CBMN assay to determine DNA damage in subjects exposed to anaesthetic gases. We also compared performance of the CBMN assay with other DNA damage assays employed and identified strengths and weaknesses of the published studies. We have retrieved 11 studies, published between 1996 and 2013, reporting MN associated with occupational exposures (operating room personnel). The individual job categories were often described (anaesthesiologists, technicians, radiologists) among cases, as well as duration of exposure. All studies reported the compounds present at the workplace and, in some instances, the exposure levels were measured. Controls were usually recruited among personnel at the hospital not exposed to anaesthetics or they were healthy unexposed subjects from general population. The number of investigated subjects, due to the character of the occupation, was relatively smaller than those investigated in other occupational monitoring settings. Overall, the majority of the studies were age- and gender- matched (or investigated only males or females) while less attention was given to lifestyle confounders. Appropriate measurement of exposure, available in approximately half of the studies only, was compromised by the lack of the personal dosimetry-based determinations.

In all studies, higher MN frequencies were observed in exposed individuals. The meta-analysis of mean MN frequency of combined studies confirmed this tendency (log mean ratio=0.56 [0.34–0.77]; $P=3.51 \times 10^{-7}$). Similar differences between the exposed and controls were also observed for other biomarkers.

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Abbreviations: BNMN cells, binucleated micronucleated cells; CAs, chromosomal aberrations; CBMN assay, cytokinesis-block micronucleus assay; N₂O, nitrous oxide; PBL, peripheral blood lymphocytes; SCE, sister-chromatid exchanges.

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1. Introduction

1.1. Anaesthetics and their health impact

Anaesthesiologists as an inseparable member of operation teams may be occupationally exposed to substantial concentrations of volatile anaesthetics. Their mutagenic and carcinogenic effects have been suggested [1–6] as they pose a potential genotoxic burden and health problems for a considerable number of individuals in various branches of medicine as well for patients. Consequently, a conclusive assessment of chromosomal damage among medical personnel is of particular importance. Presently, the most frequently used anaesthetics include sevoflurane, isoflurane, desflurane, and enflurane, all of which have gradually replaced the more toxic halothane. Earlier Jalszynski et al. [3] described isoflurane- and halothane-induced elevation of DNA damage and DNA degradation in peripheral blood lymphocytes (PBL) *in vitro*. Similarly, cellular DNA damage increased after exposure to both desflurane and halothane [5]. Together with DNA damage also a frequency of chromosomal damage has been investigated among anaesthesiologists exposed to the anaesthetic gases like sevoflurane, isoflurane, and nitrous oxide (N₂O) [1,2]. Regarding N₂O, a significant increase in micronuclei (MN) frequency in PBL was recorded among nurses exposed to this compound [7,8]. In our recent study, we quantified the association between the occupational exposure of 247 physicians and nurses exposed to volatile anaesthetics in operating rooms, and the occurrence of chromosomal aberrations (CAs) compared with sex- and age-matched controls. Interestingly, exposure to anaesthetics was associated with the frequency of total CAs (OR 3.9, 95% CI 2.7–5.8; $P < 0.0001$), chromatid-type (OR 1.8, 95% CI 1.3–2.6; $P < 0.001$) and chromosome-type (OR 3.8, 95% CI 2.5–5.6; $P < 0.0001$) aberrations [9]. This is an important observation, since the levels of CAs in PBL are considered to be predictive biomarkers of cancer risk and their elevated frequencies were found at cancer diagnosis [10–13].

1.2. Mechanisms of DNA and chromosomal damage by anaesthetics

A putative mode of action of isoflurane could be the formation of reactive trifluoroacetyl ester, resulting in modification of N7-position of guanine. However, induction of oxidative DNA damage has also been postulated for isoflurane, sevoflurane, and N₂O [14–16]. Efficient DNA repair machinery that removes arising DNA damage, comprising several distinct pathways, ensures effectively genomic integrity. Alterations in the DNA repair increase the vulnerability of the cells, resulting in an accumulation of mutations

in the genome, which may ultimately result in tumorigenesis [17]. DNA repair is closely associated with fundamental cellular processes: (a) DNA replication – its deregulation occurs through replication-blocking DNA lesions, the activation of certain oncogenes, loss of function of certain tumour suppressors, and promoting replication stress that triggers double-strand breaks (DSB) formation and leads consecutively to unscheduled recombination events and chromosomal rearrangements; (b) chromosomal segregation – defects in chromatid cohesion, spindle formation or mitotic checkpoints may lead to aberrant chromosomal segregation; (c) telomere maintenance – recognition and signalling of DNA damage is a prerequisite for the induction of subsequent cellular responses such as increased repair, cell cycle arrest and apoptosis [18–20]. Chromosomal damage, such as structural CAs, arises as consequences of direct the DNA damage or due to the replication on a damaged DNA template [21]. The lesions causing double-strand breaks are mainly responsible for chromosome-type of aberrations, whereas chromatid-type of aberrations arise as a consequence of DNA lesions generated by numerous genotoxic compounds since G₀ phase, which are insufficiently repaired prior to entering of the cell into S-phase [22]. These aberrations are considered as causative events in the malignant transformation [23] and represent also chromosomal instability, often associated with human cancers [24–26].

1.3. MN

MN, an another form of chromosome damage, mainly originate from the acentric chromosome fragments, chromatid fragments or whole chromosomes that fail to be included in the daughter nuclei at the completion of telophase during mitosis due to improper attachment to the spindle during anaphase segregation [27]. There are several mechanisms leading to this chromosomal damage, mostly associated with mis-repaired or unrepaired double-strand breaks. Another mechanisms leading to MN formation may also arise as a consequence of altered excision repair (especially with incomplete gap-filling step) or the incorporation of inappropriate bases into DNA [27].

The cytokinesis-block micronucleus (CBMN) assay is the most widely used method for measuring MN in cultured human cells, with scoring specifically restricted to once-divided binucleated cells after blocking cytokinesis with cytochalasin B. In this way, MN are scored only in cells able to divide and express the damage. The CMBN assay has become one of the standard cytogenetic tests for genetic toxicology *in vitro* and human biomonitoring of genotoxic exposures over the years while the Human Micronucleus (HUMN) project is aiming at determining the predictive value of MN as a

biomarker with the respect of the cancer risk [28] (www.humnn.org). Guidelines have been published for the standardization of the technique and scoring of slides [29].

The present review systematically investigates the association of MN in lymphocyte CBMN assay with humans exposed to the anaesthetic gases, mainly in operating rooms. The aim is to provide evidence whether or not MN in PBL cells may serve as a reliable indicator of the genotoxicity and the impairment of chromosomal integrity among subjects occupationally exposed to anaesthetics [30]. Secondary aims are: (i) to compare performance of the CBMN assay against other DNA damage biomarkers if such data are available in the studies, (ii) to identify strengths and weaknesses of the published studies and (iii) to collect the published numerical data to enable a meta-analysis of combined studies.

2. Methods

2.1. Selection of papers

In the present study, two of the authors (A.N. and L.M.) have mined all the available literature that investigated the association of MN in the lymphocyte CBMN assay with human *in vivo* exposure to the specific group of anaesthetic gases. Specifically, we have searched through PubMed by using following keywords: “micronuclei”; “cytogenetics”; “anaesthetics”; “operating room personnel” and “halothane”; “sevoflurane”; “isoflurane”. We have not considered publications in other than English language and those related to *in vitro* experiments.

Among the retrieved studies on the topic, we have also included our unpublished investigation performed on 46 anaesthesiologists occupationally exposed to volatile anaesthetics (sevoflurane/isoflurane; Table 1) in operating rooms (University Hospital in Martin, Slovakia) (Pavel Vodicka et al. personal communication). These subjects were a part of our recently published study on CA frequencies [9], so both MN and CAs were available in the same subjects. The control group (23 subjects) consisted of healthy medical staff without any apparent exposure to genotoxic agents from the same hospital. The study was approved by the Ethics committee of the Jessenius Faculty of Medicine in Martin, Slovak Republic. Exposure to volatile anaesthetics (sevoflurane and isoflurane), estimated by ambient air measurements and based on 2 measurements at 24 different places in various operating rooms at the University Hospital, was 200.3 mg/m³ (range 64.2–483.9 mg/m³).

2.2. Micronucleus assay

The technique for the determination of MN in PBL has been described previously [31]. Two thousand binucleated cells for an individual were scored from coded slides for the presence of MN and the results expressed as the frequency of BNMN cells [32]. The cytogenetic analysis for CAs frequency in PBL has recently been described [9]. Identification of the type of aberrations was performed according to the standard classification.

For each study we have recorded whether the subjects were exposed predominantly occupationally or environmentally.

Table 1
Description of the investigated studies.

Study	Year	Study population (exposed/controls)	Occupational exposure	Compounds	Concentrations at workplace	Duration of exposure (years)	Main Outcome
Chang et al. [1]	1996	18/18	operating room personnel	N ₂ O	n.d.	13.7 ± 1.2	↑
Hoerauf et al. [39]	1999	10/10	operating room personnel	N ₂ O/isoflurane	5.3 ppm isofl./12.3 ppm N ₂ O (8hTWA)	n.d.	↔
Rozgaj and Kasuba [35]	2000	28 anesthesiologists + 16 technicians/32 controls	operating room personnel	N ₂ O/halothane	n.d.	12.9 (1–32)/11.6 (1–25)	↑
Rozgaj et al. [36]	2001	43/26	operating room personnel	N ₂ O/halothane	n.d.	12.9 ± 8.5/11.9 ± 5.4/11.4 ± 9.1	↑
Wiesner et al. [33]	2001	25 High level exposed/25 controls, 25 Low level exposed/25 controls	operating room personnel	N ₂ O/halothane/isoflurane	170 ppm N ₂ O, 4 ppm others/12 ppm N ₂ O, <0.5 ppm other	Month bef. sampling: High 134 ± 31 h, Low 126 ± 37 h	↑ in High level group
Pasquini et al. [40]	2001	46/66	operating room personnel	N ₂ O/enflurane	n.d.	M: 13.4 ± 10.6 (1–37) F: 10.9 ± 5.7 (1–21)	↑ (↑ Females)
Bilban et al. [38]	2005	153 anaesthesiologists/197 (radiologists) + 153 (residents) controls	operating room personnel	N ₂ O/halothane/isoflurane	not retrievable	12.9 ± 6.5	↑
Lewńska et al. [7]	2005	46/28	operating room personnel	N ₂ O/sevoflurane/isoflurane	N ₂ O: 14–2308 mg/m ³ , others <TLV 18 mg/m ³	17.7 ± 10.1 (1–37)	↑
Wiesner et al. [4]	2008	15/15 (analysed 14/14)	operating room personnel	sevoflurane	sevofl. 0.2 ppm	n.d.	↔
Rozgaj et al. [2]	2009	50/50	operating room personnel	N ₂ O/sevoflurane/isoflurane	n.d.	13.0 ± 9.0	↑
Araujo et al. [8]	2013	30/30	operating room personnel	N ₂ O/sevoflurane/isoflurane/enflurane/halothane	n.d.	13.8 ± 10.9 (2–37)	↑
Vodicka et al. Personal communication ^a	2015	46/23	operating room personnel	N ₂ O/sevoflurane/isoflurane	200.3 mg/m ³ (range 64.2–483.9)	12.8 ± 9.6	↑

n.d. not determined, ↑ increased MN frequency in exposed subjects, ↔ MN moderately higher but non-significant.

^a For details Musak et al. [9].

Whenever available we have collected all information about: (1) specific compounds present in the environment of operating rooms; (2) concentrations of single compounds detected in the above area; and (3) duration of exposure of investigated subjects.

All extractable information about CMBN assay was also included (number of cells scored, MN frequencies, fold-changes in comparison with healthy subjects, significances of the results etc.). All available population characteristics were also recorded (age, gender, smoking habit, alcohol consumption).

2.3. Quality score

Two of the authors (P.V and L.V) have classified the identified studies according to their quality by using a methodological quality assessment scale, which included nine items: number of subject (I) in control group and (II) in exposed group (1 = < 20, 2 = 21–50, 3 = > 50); (III) age-, (IV) gender-, (V) smoking- and (VI) alcohol-intake matching (1 = no data available or significant difference between groups, 2 = partially matched, 3 = perfectly matched); (VII) nutritional intake matching (1 = not matched, 2 = matched using a food intake questionnaire, 3 = matched using blood nutrient measurements); (VIII) exposure to anaesthetic gases (1 = assessed by questionnaire only, 2 = assessment by measurement in ambient environment, 3 = assessment by measurement in body fluids); (IX) number of binucleated cells scored per subject (1 = < 1000, 2 = 1000–2000, 3 = > 2000). The quality scores ranged between 9 and 27.

2.4. Other analysed endpoints

Additionally we have searched for other DNA/chromosomal damage assays investigated (e.g. buccal MN assay, comet assay, CAs, sister-chromatid exchanges (SCE) etc.) and reported in the

identified studies. If yes, data were also collected and compared with MN results.

2.5. Statistical analysis

Mean MN frequencies and standard deviations (SD) were collected from the studies. Two studies reported median MN frequency and range as summary statistics [4,33]. In these cases, mean MN frequency and SD were estimated according to [34]. Results from single studies were meta-analysed using both fixed and random effects models, and represented by Forest plots. A sensitivity analysis was performed, excluding the studies one at the time and repeating the meta-analysis. The Egger's test was used to assess publication bias (see Funnel plot).

The statistical analyses were done by using the *rmeta* package from the free software environment for statistical computing R.

3. Results

By mining all literature for studies investigating the association between MN frequencies by the lymphocyte CBMN assay and human *in vivo* exposure specifically to anaesthetics, we have retrieved 11 studies, all in English language, published between 1996 and 2013. Two studies [35,36] were most likely the continuation/expansion of the first study by the same first author [35]. However, they were kept separated in the reporting tables, as individual studies. One extra study, very similar and further commented upon but not included in the analyzed publications, was performed on MN in buccal cells [37]. All retrieved studies are summarized and briefly described in Table 1.

There was a substantial difference among studies regarding the size of the study population, comprising exposed workers and unexposed subjects. Few of them (25%) investigated less than

Table 2
Evaluation and scoring of the studies.

Study	Year	BNMN scored/subject	Fold changes in MN (exposed/controls)	Age-matching	Gender distribution	Gender-ratio (females/males)	Smoking status-matching	Appropriate measurement of chemical exposure	Quality score
Chang et al. [1]	1996	1000	3.83	yes	only females	n.a.	only smokers	no	16
Hoerauf et al. [39]	1999	2000	1.28	yes	exposed 6f/4m, controls matched	E:1.5, C = same	only smokers	yes	17
Rozgaj and Kasuba [35]	2000	1000	4.34/3.59	no	anaesthesiol. 14/14, technicians 8/8 controls 15m/17f	E-anest:1.0, E-techn:1.0, C:1.1	no	no	13
Rozgaj et al. [36]	2001	1000	n.a.	no	exposed m27/26f, controls 15/11	E:0.97, C:0.73	no	no	13
Wiesner et al. [33]	2001	2000	HIGH 1.24/LOW 0.93	yes	HIGH: exposed 4m/21f, controls 1m/24f; LOW: exposed 14m/11f controls 16m/9f (matched)	HIGH E:5.5, C:24, LOW E:0.9C:0.6, combined E:1.7C:1.9	yes	yes	18
Pasquini et al. [40]	2001	2000	1.19	no	cases 22f/24m, controls 31f/35m	E:0.91, C:0.89	no	no	14
Bilban et al. [38]	2005	500	1.15/1.19	yes	only females	n.a.	no	yes	17
Lewińska et al. [7]	2005	1000	2.07	n.a.	only females	n.a.	no	yes	15
Wiesner et al. [4]	2008	2000	1.12	yes	exposed 8m/6f, controls 9m/5f	E:0.75, C:0.56	no	yes	14
Rozgaj et al. [2]	2009	1000	2.35	yes	exposed: 12m/38f, controls matched	E:3.2, C = same	yes	no	18
Araujo et al. [8]	2013	2000	2.00	yes	exposed 14m/16f, controls matched	E:1.1, C = same	no	no	16
Vodicka et al. Personal communication ^a	2016	2000	1.38	yes	exposed 9m/37f, controls 5m/18f (matched)	E: 4.1, C 3.6	no ^b	partially (approximate estimation)	17

n.a. not applicable; E: exposed, C: controls.

^a Score calculation is reported in Section 2 (Methods).

^a For details Musak et al. [9].

^b More smokers among exposed.

20 subjects per category. Only one study exceeded 150 subjects per category [38]. Approximately in two thirds of studies the size of unexposed group exceeded that of exposed subjects.

All studies investigated MN associated with occupational exposures (operating room personnel). In some cases the individual job categories were described (anaesthesiologists, technicians, radiologists) among cases. Controls were recruited either among personnel of the hospital not exposed to anaesthetics or other healthy unexposed subjects.

The studies reported the compounds present at the workplace: N₂O was reported in all studies, followed by sevoflurane, isoflurane, halothane and enflurane. However, only five studies clearly reported any details about the exposure levels ([4,7,33,39] Vodicka et al. unpublished data).

Except for 2 studies [4,39], all other investigations also reported details about the duration of exposure of the enrolled subjects. In all cases the exposure exceeded ten years. Wiesner et al. [33] in one of the studies reported the exposure in hours within the last month. In six studies 2000, in five 1000 and in one 500 BNMN cells per subject [38] were scored.

In all studies a consistent increase in MN frequencies was observed in cases in comparison with controls. However, in 2 studies the difference did not reach the significance [4,39], in one MN frequency was only significant among females [40] and in another one only in highly exposed subjects [33].

In 11 out of 12 studies the fold- changes in MN frequency between the exposed and control individuals could have been calculated (Table 2). Except for the low exposed individuals reported by Wiesner et al. [33], where MN frequency was even lower than those in controls (fold-change 0.93), the remaining studies showed a fold-change ranging between 1.12 and 4.34. Interestingly, more recent studies (since 2005) show more consistent values of MN frequency fold- change, ranging between 1.12 and 2.35.

Table 2 also reports all information retrieved to build a “quality” score of the investigated studies, taking into the account the study design, measurement of the selected exposure and the potential effect of confounders. Analysis of the quality score revealed that median score was 16 with a minimum 13 and maximum 18. The studies failed short from the maximal score 27, since details on alcohol consumption and nutritional intake were not satisfactorily reported, and any statistical analyses considered these factors. An additional inherent factor, devaluating the total score, is the size of

the operating theatre teams. Majority of the studies were age-matched and took care about gender-matching, with some studies performed only among females. Less attention was given to the smoking habit. Measurement of an exposure, available in 5 studies only, was compromised by the lack of any determinations in body fluids.

Several of the studies employed other tests besides lymphocyte CBMN assay to address chromosomal/DNA damage. Table 3 lists all other investigated endpoints and their main outcomes, including a comparison of the outcomes of these analyses with the CBMN assay. Except for two studies, all other were focused on endpoints, such as CAs and SCEs. One study investigated DNA damage by comet assay, one MN by FISH and another CBMN implemented with polymerase inhibitor cytosine arabinoside (ARA-C). In general, there was overall a good agreement between other tests and CBMN assay outcomes.

Statistical analysis for evaluating the relationship between CBMN and exposure to anaesthetic gases has been prevented by the lack of studies with the quantitative exposure data. In fact, anaesthetic's exposure was determined in 5 out of 12 existing studies, however in all instances the measured concentrations represent only semi quantitative parameter, since they have not been monitored by personal dosimeters.

We finally tested statistically, via meta-analysis of all retrieved data reported in the analysed studies, whether MN frequency in lymphocytes measured using the CBMN assay was associated with the chemical exposure. A fixed effect model test for heterogeneity indicated the adequacy of a random effects model for this meta-analysis, which was finally used to estimate the mean difference in MN frequency between exposed (n = 504) and unexposed individuals (n = 674) combining all studies; this difference resulted statistically significant (log mean ratio = 0.56 (0.34–0.77), P = 3.51×10^{-7} ; Fig. 1). Interestingly, one study [33] that reported higher median MN frequency in exposed subjects, when recalculated for mean frequency, showed a lower frequency in comparison with unexposed controls.

Gender, age and smoking status that might affect MN frequencies were not stratified in this meta-analysis, since not all investigators reported uniformly information for MN frequencies according to these factors. However, several of the studies, as previously stated were age- or gender-matched (or performed only in females).

Table 3
Comparison of MN outcomes with those of other investigated tests.

Study	Year	Parameters investigated	Main outcome	Fold changes (cases/controls)	Concordance results among different tests
Chang et al. [1]	1996	CBMNA (ARAC)	↑	2.83	yes
Hoerauf et al. [39]	1999	SCE	higher but non-significant	1.37	yes
Rozgaj and Kasuba [35]	2000	CAs	↑ acentric fragments CA-anesthesiologists, ↑ dicentric chrom CA-technicians	n.d.	slightly yes
Rozgaj et al. [36]	2001	CAs, SCE	↑ CA, ↑ non-significant SCE	n.d.	yes
Wiesner et al. [33]	2001	n.a.	n.a.	n.a.	n.a.
Pasquini et al. [40]	2001	SCE	↓ non-significant	0.88	no
Bilban M. et al. [38]	2005	CAs, SCE	↑ both	CAs: 1.19/1.03, SCE: 1.19/1.03	yes
Lewińska et al. [7]	2005	MN FISH	↑	1.78 (MN+ 49/MN-51 vs. MN+ 43/MN- 57)	yes
Wiesner et al. [4]	2008	SCE	↑	1.29	yes, but stronger significance
Rozgaj et al. [2]	2009	Comet assay	↑ TL, non sig. TM	TL: 1.20/TM: 1.14	yes
Araujo et al. [8]	2013	n.a.	n.a.	n.a.	n.a.
Vodicka et al. Personal communication ^a	2015	CAs	↑	1.85	yes ^b

↑: increased levels; n.a. not applicable. CA: chromosome aberrations, MN FISH: fluorescence in situ hybridization (MN+: micronuclei with centromere signal, MN- with no centromere signal), SCE: sister chromatid exchanges. For Comet assay: TL, tail length; TM, tail length.

^a R = 0.237, P = 0.048.

^b For details Musak et al. [9].

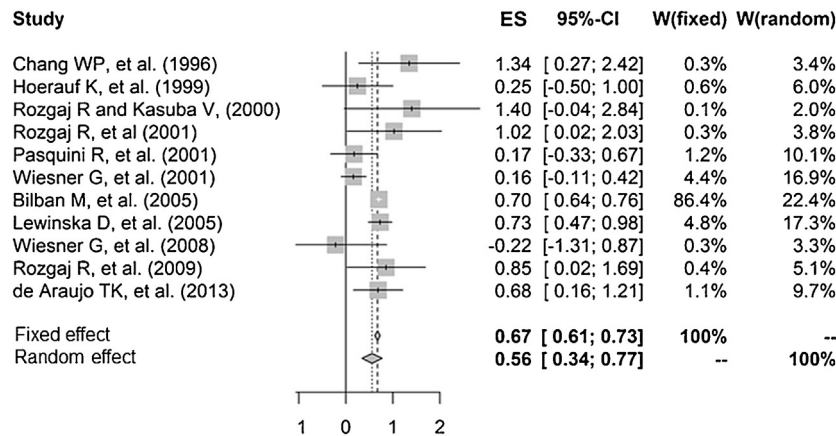


Fig. 1. Forest plot reporting the mean MN frequency in occupationally exposed subjects to anaesthetics and unexposed controls via the meta-analysis of all studies retrieved by the literature.

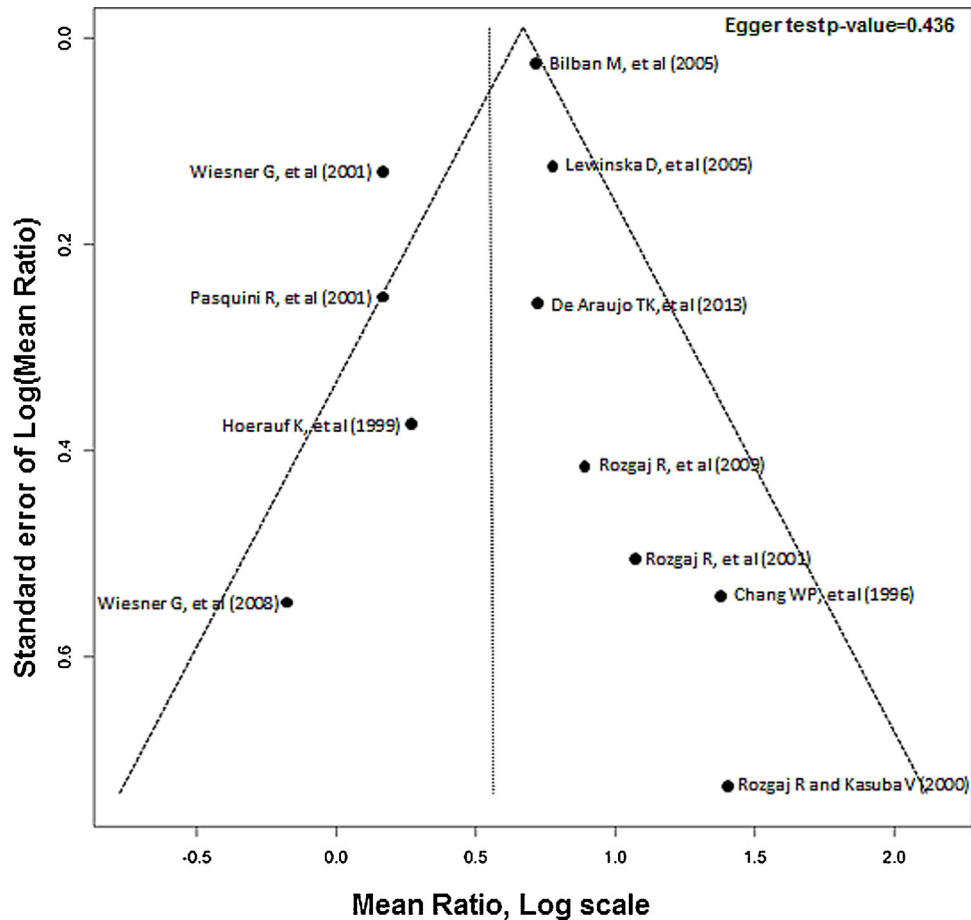


Fig. 2. Funnel plot for publication bias.

Using Egger's test, no publication bias was found ($P=0.436$), as shown by lack of asymmetry by the Funnel plot (Fig. 2). From the sensitivity test, no individual study was found to significantly impact the overall results, with the exception of [33]. Despite the 95% CI of this study, its removal reduced the heterogeneity and increased the significance of the association (Supplementary Table 1).

4. Discussion

Chromosomal damage in PBL (such as MN and CA) represent transient markers in process of carcinogenesis. Both are associated with failed or incomplete DNA repair (predominantly DSB, but excision repair as well). In particular, MN originate mainly from the

acentric chromosome fragments, chromatid fragments or whole chromosomes that fail to be included in the daughter nuclei at the completion of telophase during mitosis due to improper attachment to the spindle during anaphase segregation [41]. The relationship between the frequency of MN and exposure to genotoxic occupational and environmental agents has been sufficiently proven [41,42]. The importance of MN has most recently been highlighted by genome sequencing that has uncovered a new mutational phenomenon in cancer and congenital disorders called chromothripsis. Chromothripsis is characterized by extensive genomic rearrangements and an oscillating pattern of DNA copy number levels, all restricted to one or a few chromosomes. The tentative mechanism of chromothripsis is based on the anticipation of the physical isolation of chromosomes in aberrant nuclear structures, micronuclei. The MN formation may indeed generate a spectrum of genomic rearrangements, some of which recapitulate all known features of chromothripsis, as proven by live cell imaging and single-cell genome sequencing. These events are restricted to the mis-segregated chromosome and occur within one cell division. The mechanism for chromothripsis can involve the fragmentation and subsequent reassembly of a single chromatid from a micronucleus [43]. The genetic and mechanistic background of MN and CA frequencies in human peripheral lymphocytes is slowly emerging [44–47] and confrontation of *in vivo* DNA repair characterization with the kinetic of chromosomal damage is inevitable.

By reviewing currently available literature on MN (an indicator of genotoxicity and the impairment of chromosomal integrity) in the lymphocyte CBMN assay with human *in vivo* exposure specifically to anaesthetics we have retrieved 11 relevant studies and one consistent set of data not yet published. The size of the study population seldom exceeded 50 exposed subjects, which reflects limited number of employees in operation teams. In all studies the analysed MN were associated with occupational exposures among operating room personnel: N₂O was reported in all studies, followed by sevoflurane, isoflurane, halothane and enflurane. Despite five studies reported some details about the exposure levels, these may be considered as semi-quantitative (no individual data based on personal dosimetry available). In some instances the individual job categories were clearly articulated (anaesthesiologists, technicians, radiologists) among cases, however the risk assessment for a particular profession was not possible. Hereby, we have analysed each available study for study design, important technical aspects (number of BNMN cells per subject scored) and the availability of an important confounder and population data (e.g. duration of exposure, smoking, sex and age matching). Emerging arbitrary quality score of the investigated studies revealed that none of the studies was satisfactorily conducted (failing short from the maximal score 27), since details on alcohol consumption and nutrition were not comprehensively reported and evaluated in any report. An inherent factor contributing to the decreased score is the size of the operating teams. Measurement of exposure, available in 5 studies only, was compromised by the lack of any determinations in body fluids.

The vast majority of studies revealed the fold- change in CBMN ranging between 1.12 and 4.34. Interestingly, more recent studies (since 2005) show more consistent values of MN frequency fold-change (range between 1.12 and 2.35).

Finally the meta-analysis clearly showed a statistically significant different mean MN frequency between exposed workers and unexposed controls.

Thus, several different evidences point to the biological relevance of anaesthetic gases in triggering chromosomal damage.

Ten studies employed other tests besides lymphocyte CBMN assay to address chromosomal/DNA damage, such as CAs and SCEs,

DNA damage assessed by means of comet assay, and MN by FISH. Only one study investigated CBMN implemented with polymerase inhibitor cytosine arabinoside (ARA-C). The overall good agreement between other tests and CBMN assay outcomes further underlines biological effects of anaesthetic gases in humans.

Our previous report on 247 medical workers occupationally exposed to anaesthetics gases [9] documented higher CA frequencies in the exposed subjects. A subset of 46 individuals exposed to anaesthetics and 23 control individuals were assayed also for the MN frequency. MN were significantly higher in the exposed individuals than in control ($5.23 \pm 2.77\%$ vs. $3.78 \pm 2.60\%$; Mann-Whitney *U*-test, $P=0.012$). The non-parametric Spearman analysis revealed significant correlations between the frequency of MN and age ($R=0.272$, $P=0.023$), and female gender ($R=0.332$, $P=0.005$). Due to the limited number of measurements MN frequencies were not stratified according to smoking status and job category. A significant associations with exposure to anaesthetics ($R=0.271$, $P=0.023$) and frequencies of total CAs ($R=0.237$, $P=0.048$) add additional evidence to the genotoxic effects of anaesthetics (Vodicka et al., *Personal communication*). Our earlier experience also suggested the association of the MN frequency with parameters of the occupational styrene exposure, particularly with a cumulative one. However, the links between the extent of DNA damage, DNA repair rates –reflecting base-excision repair (BER)- and MN formation remained inconclusive [32]. In our more recent report we failed to prove mechanistic links between MN frequencies, BER rates or mRNA transcripts of the prominent BER genes [48].

Further search for mechanistic implications, especially with respect to DNA repair, is underlined by the important issue encountered by researchers employing CBMN assay, whether the formation of MN occurs *in vivo* or MN are formed *ex vivo* during cell culturing [49]. In another review, Speit points out an insensitivity of the CBMN assay in the absence of araC, an inhibitor of DNA repair synthesis, and postulates that positive results after *in vivo* exposure to agents inducing excision-repairable lesions cannot easily be explained in human biomonitoring studies [50]. However, our experience based on measuring MN from animal bone marrow (without the stimulation and cytochalasin B block) indicates that an inhalation exposure of 1,3-butadiene (BD) resulted in dramatic increase in MN formation in earlier time intervals of the 28 days exposure, followed by gradual decrease almost to control levels 28 days post-exposure. In this experiment the relationships between MN and either DNA damage or BER capacity were not pronounced, most likely due to the determination of these parameters in different tissues [51]. On the other hand, the kinetic of MN accumulation and disappearance during the BD exposure and in post-exposure period does not support commentary of [49], who assumed that the genotoxic damage is being entirely repaired or drives the cell to apoptosis. On the contrary to BD, which seems to be clastogenic, only moderate effect was observed after subacute inhalation exposure to styrene in mice, but, interestingly, a correlation between MN and DNA single-strand breaks were observed in bone marrow, suggesting mechanistic link between these two parameters [52].

5. Conclusions

In general, results from CBMN assay well reflect genotoxicity of anaesthetic gases and this observation is further supported by other assays, too. However, the majority of investigations is based on small size study populations and the study design is often not properly taking care of exposure measurement or the effect of potential confounders.

There are several open mechanistic issues, such as the kinetic of DNA damage formation, DNA repair activity, kinetic and genetics of

chromosomal segregation (defects in chromatid cohesion, spindle formation or mitotic checkpoints). Mode of action of specific xenobiotics should be taken into consideration (differences in their clastogenicity), including their biotransformation, bioavailability and possibility of cumulative effect over time. Finally, we shall ask whether the data in surrogate (PBL) tissue correspond to the situation in target tissue (where malignancy may be expected), as tissue differences in DNA repair have been suggested [53].

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.mrrev.2016.04.003>.

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